MIL-P-82013C
5 January 1983
SUPERSEDING
MIL-P-82013B
12 May 1976

MILITARY SPECIFICATION

# PLATFORM, PIPELINE LAUNCHING AND RETRIEVING: SUBMARINE FUEL LINES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers an interlocking series of platform-mounted pipe rollers with fittings and anchoring provisions.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-P-24441 - Paint, Epoxy Polyamide.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3835

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

BUREAU OF YARDS AND DOCKS (NAVAL FACILITIES ENGINEERING COMMAND)

6028019 - Launching Assembly Submarine Fuel Lines Details Launching Rollers and Platform.
6028019 - Launching Assembly Submarine Fuel Lines Details Make Up Rollers and Trailer.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN PETROLEUM INSTITUTE (API)

Std 5A - Specification for Casing, Tubing, and Drill Pipe.

(Application for copies should be addressed to the American Petroleum Institute, 1801 K Street, N.W., Washington, DC 20006.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Specification for Structural Steel.
A283 - Specification for Low and Intermediate Tensile Strength Carbon Steel Plates of Structural Quality.
A675 - Specifiation for Carbon Steel Bars Subject to Mechanical Property Requirements.
B584 - Specification for Copper Alloy Sand Castings for General Applications.
E18 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)



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AMERICAN WELDING SOCIETY (AWS)

AWS D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 N. W., 7th Street, Miami, FL 33125.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 Description. Each submarine fuel line makeup rollers, launching rollers, and platform shall include all prefabricated members, parts, materials, and fasteners shown and specified on Drawing 6028019.

3.2 Standard commercial components. The materials and parts of the pipeline launching and retrieving platform assembly shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial components. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial components, shall be included in the pipeline launching and retrieving platform being furnished. Standard commercial components are components which have been sold or are being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.2), the contractor shall furnish a pipeline launching and retrieving platform assembly for first article inspection and approval (see 4.2.1 and 6.4).

3.4 Identical items. All units of the same classification furnished under a specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviations will be acceptable without written approval of the contracting officer.

* 3.5 Material. Material shall be as specified herein. Materials shall be free from all defects and imperfections that might affect the serviceability



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and appearance of the finished product. Bushing in bearings shall be of high-leaded tin bronze, conforming to ASTM B584, copper alloy No. 932. Steel compositions shall conform to SAE J403. Rollers, pins, and thrust washers shall be of steel conforming to SAE J414, composition 1045, and shall be surface-hardened to a Rockwell hardness between C55 and C60. Steel for launching and make-up rollers, pin support angles, platforms, and platform anchors shall conform to SAE J414, composition 1020 cold drawn; ASTM A36; grade A; ASTM A283, grade D; or ASTM A675, grade 60. Anchor rods shall be of steel conforming to SAE J414, composition 1042, cold drawn. The dog locking device on roller assemblies shall be fabricated from a cut length of steel mechanical tubing conforming to SAE J414, composition 1015, cold drawn, welded to a rectangular shoe of steel conforming to SAE J414, composition 1015, cold drawn, with a commercial wrought steel washer continuously welded about the mechanical tubing. The dog wrench shall be of steel conforming to SAE J414, composition 1045.

* 3.6 Design, construction, and performance. When placed end-to-end on approximately level ground, such as a beach, the platforms shall fit snugly end-to-end in a straight line. Make-up rollers, spaced at 10-foot intervals, shall not be damaged by the rotation of 6.625 inch nominal diameter API Std 5A, grade J, 120 pounds per foot, buttress thread and coupled steel oil well casing pipeline sections when being made up into an extended length of pipeline by use of a powered pipe coupling tong. The make-up roller platform shall also aline for welding random lengths of 8 inch and 12 inch nominal outside diameter seamless welded steel pipeline sections during routine pipeline fabrication. Launching rollers, when alined on platforms and connected end-to-end to form a continuous string not less than 150 foot long, shall not be damaged and misalined after transporting a length of 240 foot, or greater, of 4 inches through 12 inches nominal sized pipeline at a speed of not less than 130 feet per minute. After lubrication, rollers shall be rotatable by the pressure of one hand. In the event of a conflict between the drawings and the specifications, the specifications shall govern. When specified (see 6.2), a dog wrench shall be furnished.

* 3.6.1 Sleeve and dog assembly. The wrought steel washer of the sleeve and dog assembly shall be welded in place on the mechanical tubing so that after insertion of the assembly through the base plate of either roller and the rectangular hole in the platform, a quarter turn of the dog wrench will secure the connecting items in working position.

3.6.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534, straight type, thread A, PTF Special Extra Short.

3.6.3 Pipe plugs. Pipe plugs shall be steel, with square heads and screw threads identical with those of the lubrication fittings (see 3.6.2).

* 3.7 Lubrication. Rollers shall be lubricated prior to delivery with lubricants designated for use in the 0° Fahrenheit (F), to 32°F temperature range, and tagged to identify the type and temperature rating of the lubricant used. After lubrication, the fittings shall be removed, added to the extra lubrication fittings, and replaced with pipe plugs.



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3.8 Cleaning and painting.

* 3.8.1 Cleaning. All steel surfaces of the unit to be painted shall be cleaned by blasting. Blast cleaning shall be to a near white metal blast or cleaner. Near white metal blast is described as a surface with 95 percent of each square inch free of all visible residue with the remaining 5 percent of each square inch having no more than very light shadows or streaks caused by rust stains or mill scale oxides. Blast cleaning shall utilize either dry sand or crushed grit made of cast iron, steel, or synthetic grit. Grit shall be of a maximum particle size no larger than that which is able to pass through a 16 mesh screen.

* 3.8.2 Epoxy-polyamide painting. Epoxy-polyamide paint shall conform to MIL-P-23331. Formulations of MIL-P-24441 shall be applied only to dry surfaces in an atmosphere having a relative humidity not greater than 80 percent and a temperature of 70°F or higher. When ambient atmosphere does not meet these requirements, all painting shall be accomplished in a humidity and temperature controlled environment. There shall be 16 hours of curing between each coat. If more than 7 days elapse between epoxy coats, the surface shall be cleaned with water and detergent, rinsed with fresh water, dried, and then a tack coat (1 to 2 mils wet film thickness) of the last coat applied before application of the next full coat called for. The two components of all the MIL-P-24441 coatings should be mixed in equal volume by first thoroughly stirring each component separately and then stirring the components together. After mixing, there shall be a waiting period of approximately 2 hours at 50° to 60°F, 1 to 1/2 hour at 60° to 70°F, and 1/2 to 1 hour above 70°F before applying the coating to insure complete curing. If spray or roller painting is utilized, all corners and welds shall first be thoroughly brush coated for each coat of paint. The paint application shall be uniform in thickness and color, smooth in appearance, free of dry over-spray, pinholes, orange peel, sags, or other film defects. Threaded surfaces shall not be painted. All painted areas shall be touched-up where paint is removed as a result of handling or assembly at the manufacturer's plant.

* 3.8.2.1 Primer and intermediate coat. Primer and intermediate coats shall consist of epoxy-polyamide paint conforming to MIL-P-24111. The primer coat shall be green, formula 150, and shall have a minimum 2.5 mil dry film thickness. The intermediate coat shall be white, formula 152, and shall have a minimum 2.5 mil dry film thickness. Priming of the cleaned surfaces shall be completed within a maximum of 4 hours after blast cleaning. Epoxy-polyamide paint shall be applied in accordance with 3.8.2.

* 3.8.2.2 Finish coat. All metal surfaces to be painted shall be given one finish coat of epoxy-polyamide paint conforming to MIL-P-24441. The finish coat shall be as specified (see 6.2), haze gray No. 27, formula 151, and shall have a minimum 3.0 mil dry film thickness.

* 3.9 Identification marking. All rollers, platforms, and platform anchors shall be marked for identification in accordance with MIL-STD-130.

* 3.10 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the



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finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.11 Welding. Welding procedures shall be in accordance with a AWS D1.1 welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.12 Workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

* 4.2.1 First article inspection. The first article inspection shall be performed on one unit when a first article is required (see 3.3 and 6.2). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

* 4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.5, the tests of 4.6, and the packaging inspection of 4.7. This inspection shall be performed on the samples selected in accordance with 4.4.

4.3 Inspection lot. All platforms and parts of like description offered to the Government at one time shall be considered a lot for purposes of inspection.



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4.4 Sampling. A random sample of platforms and parts shall be selected from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. Examination of the production platforms and parts shall be based on inspection level II, and an Acceptable Quality level (AQL) of 4.0 defects per one hundred units.

* 4.4.2 Sampling for tests. Tests of the production platform and parts shall be based on inspection level S-4 and an AQL of 2.5 defects per one hundred units.

* 4.5 Examination. Each unit selected shall be examined for complicance with the requirements specified in Section 3 of this specification. Corrections required, following failure to meet specified requirements, shall receive particular attention of adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensionsal measurements, with particular emphasis on interchangeability of like parts. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6 Tests. Units selected shall be tested in accordance with 4.6.1. Dimensional nonconformance, evidence of warp, inability to rotate any lubricated roller by pressure of one hand, poor fit, and faulty operation of the sleeve and dog assembly shall constitute defects for each unit and all interconnecting parts. Hardness less than C55 and greater than C60 shall result in hardness testing of all other corresponding items included under the contract. Surface roughness and irregularties that result from the tests, shall be polished from the surface of all accepted items and parts.

* 4.6.1 Production parts tests. Prior to assembly, the rollers, roller pins, and thrust washers shall be tested in not less than three plases each for Rockwell hardness in accordance with ASTM E18. Hardness less than C55 and greater than C60 shall constitute a defect, and shall result in hardness testing of all other corresponding items included under the contract. Surface roughness and irregularities that result from the tests, shall be polished from the surface of all accepted items and parts.

4.7 Packaging inspection. The preservation, packing, and marking of the the platform assemblies shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.



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5.1.1.2 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows: Shall be coated with type P-1 preservative.

5.1.1.3 Fittings. The loose lubrication fittings and pipe plugs required for each complete platform assembly shall be preserved method IC-1 and packaged together in a box conforming to PPP-B-636, class weather-resistant. The fiberboard box shall be closed in accordance with method V of the appendix to the box specification when it is to be shipped as the exterior container.

5.1.2 Commercial. Material shall be packaged in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A.

* 5.2.1.1 Rollers, anchors, rods, wrenches, and fittings. The launching rollers, make up rollers, platform anchors, anchor rods, dog wrenches, and fittings shall be packed in close-fitting boxes conforming to PPP-B-636 cushioned, blocked, and braced to prevent movement within the boxes.

* 5.2.1.2 Platforms. The platforms shall be bundled into secured lifts with flat steel strapping or bolting in combination with suitable wood blocking or battens, as required to form compact nonshifting bundles. The strapping shall be 0.035 by 1-1/4 inch flat steel conforming to QQ-S-781, class 1, type I, finish B. Strapping shall be spaced not to exceed approximately 36 inches on center. Edge protectors of metal or pressed fiberboard shall be used where strapping bears on metal or pressed fiberboard shall be used where strapping bears on sharp metal edges. The strapping shall be stapled to any wood blocking or battens provided.

5.2.2 Level B. The platform assemblies shall be packed as specified for level A, except that the boxes shall be domestic type or class 1, as applicable, and strapping be other than galvanized.

5.2.3 Commercial. Material shall be packed in accordance with MIL-STD-1188.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use. Pipeline launching and retrieving roller platform assemblages are used at shoreline sites to facilitate the making up, breaking down, playing out, and pulling in submarine bottom-laid construction of fuel storage tank farms, and during the make up and installation of 4 inch through 12 inch nominal outside diameter pipelines.



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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval see 3.3, 4.2.1, and 6.4.
c. When a dog wrench is required (see 3.6).
d. When finish color coat is specified (see 3.8.2.2).
e. Level of preservation and level of packing required see 5.1 and 5.2.

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the unit will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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